

ANALYSTLAB AFRICA

Experience Lab Internship Programme

Week 2: FinTrust Digital Bank — Analyse & Prepare

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Part A: Data Preparation & Quality Assessment

1. Number of Records and Columns

Dataset	Records	Columns
Customers	1,500	12
Transaction	12,000	11

2. Missing Values

The Customer dataset contains no missing values across all 12 columns.

The Transaction dataset contains missing values in two fields:

Field	Missing Records	Percentage
Device_Type	96	0.8%
Location	96	0.8%

3. Duplicate Records

No duplicate records were identified in either dataset.

The Customer dataset also has unique Customer_ID values, with 1,500 unique customer IDs across 1,500 customer records. The Transaction dataset also contained no duplicate records based on the checks performed.

4. Incorrect or Inconsistent Values

The categorical fields were reviewed for inconsistent or unexpected values. No obvious inconsistencies were identified in the categorical variables.

The values in fields such as transaction type, channel, transaction status, customer segment and risk review flag were consistent with the expected categories provided in the data dictionary.

5. Data Type Issues

The data types were reviewed against the FinTrust data dictionary.

The main issue identified was:

Transaction_DateTime

In Excel, this field should be stored and recognised as a Date/Time value so that it can be used correctly for time-based analysis.

The field was converted to the appropriate date/time format during the data preparation process.

The other fields were generally consistent with their expected uses, including numerical fields such as Age, Tenure_Months, Digital_Engagement_Score and Amount_NGN.

6. Unusual Values

Several numerical fields were checked for values outside their expected ranges.

- Customer Age:

Customer ages ranged from 18 to 65 years.

No unusual age values were identified.

- Digital Engagement Score

The Digital Engagement Score remained within the stated 10 to 100 range.

No values outside the expected range were identified.

- Tenure

No negative values were found in Tenure_Months.

- Transaction Amount

Amount_NGN contained no negative or zero transaction amounts.

The observed transaction amount range was approximately: NGN 100.20 to NGN 636,603.06

This indicates that all transaction amounts were positive.

7. Potential Outliers

The Interquartile Range method was used to identify potential outliers in transaction amounts.

The results were:

Measure	Value
Q1	2,192.22
Q3	48,484.48
IQR	46,292.26
Lower Bound	-67,246.16
Upper Bound	117,922.86
Potential Outliers	1,474

There were 1,474 transaction amounts above the upper IQR boundary and therefore identified as potential outliers.

However, these values should not automatically be treated as errors. Since the dataset contains financial transactions, larger transactions may represent genuine high value activity. They should therefore be retained for analysis and investigated further rather than removed without business justification.

8. Relationship Between Customer and Transaction Data

The two datasets are connected through Customer_ID

The intended relationship is:

One Customer → Many Transactions

The relationship was validated by checking the Customer IDs in the transaction data.

There 0 transaction Customer IDs not found in the Customer dataset. This indicates that all transaction records can be linked to an existing customer record.

The Customer dataset contains 1,500 unique Customer IDs, confirming that Customer_ID can serve as the customer key.

9. Issues That Could Affect Analysis

The following issues should be considered during the analysis:

- Missing Device and Location data

There are 96 missing values in both Device_Type and Location. This represents 0.8% of transaction records and could affect analysis involving device or location.

- Transaction Date/Time format

The Transaction_DateTime field needed to be treated as a proper Date/Time field in Excel to support accurate time-based analysis.

- Potential high value transaction outliers

1,474 transactions were identified as potential outliers using the IQR method. These should be investigated but not automatically removed because high transaction values may be legitimate.

- No duplicate records

No duplicate records were identified, reducing concerns about double counting transactions or customers.

- Valid customer relationship

All transaction Customer IDs matched records in the Customer dataset, meaning there were no unmatched transaction records identified during the relationship check.